



P16

Typed Finality: Upgrading the Binary BFT-lite Stop Decision into `strong_stop` / `weak_stop` / `abort` Three Levels of Semantic Commit

类型化最终性：BFT-lite 停止决策升级

Typed Finality: Upgrading the Binary BFT-lite Stop Decision into strong_stop / weak_stop / abort Three Levels of Semantic Commit

Working Title (EN): Typed Finality: Upgrading the Binary BFT-lite Stop Decision into Three Levels of Semantic Commit for LLM-Agent Committees

Consensus governance paper P16 • semantic upgrade of v7.5.0 P2 "BFT-lite for Multi-Agent Stop Decisions"
 • sister paper to P26 (weighted BFT) and P32 (rules over learning) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: LLM committees produce stochastic natural language each round and cannot, like classical BFT, agree on "exactly identical bytes"; the binary STOP/CONTINUE of v7.5.0 P2 compresses "how semantically aligned" into one switch, which is too coarse. This paper upgrades it into three levels of **typed finality**—strong_stop ($\geq 2f+1$ members agree on the semantic core), weak_stop (agree only on the verdict label), abort (not aligned, turn to view change)—and takes the self-withdrawn negative result of H-CSC v2 as the red line for "what not to do." The three-level upgrade is a **design proposal • pending implementation**, with no new experimental numbers.

Evidence scope (stated only once in the whole paper): The external dialogue object of this paper is H-CSC (arXiv:2606.07316, **note v2 has been retitled** to Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides); its existence and the "typed commit objects / $2f+1$ certificate envelope" have been verified; its negative-result numbers (aggregation distance, honest retention rate, lexical predicate and learned-encoder AUROC) are **paper-reported, not independently rechecked**, and are the v2 actively-withdrawn scope. All UDOS-side numbers go back to v7.5.0 P2 (udos7/topology/consensus.py, evidence grade cpu-proto / verified). This paper newly creates no UDOS measured numbers; the benefits of three-level finality are all marked [RESULT NEEDED]. Verification date 2026-09-21.

Abstract

v7.5.0 P2 reduces the terminal decision of "whether a work order is done or needs rework" to a minimal Byzantine fault-tolerant committee: committee size $\geq 3f+1$, quorum $\geq 2f+1$, same-round equivocation invalidates the whole round, and more than f silence returns NO_QUORUM and triggers view change. At $n=7, f=2$, with 10 property tests (verified), it guards the three safety/liveness properties of "honest supermajority stops, Byzantine cannot force a stop, silence turns to a view." But it has an unstated coarseness: **the decision space is binary**—there is no intermediate state between STOP and CONTINUE, and the real semantic-consensus gradient of "everyone's conclusions are roughly right but wording diverges" is hard-compressed into the two bits of "stop or not."

H-CSC (arXiv:2606.07316, v2) precisely addresses this coarseness. It points out: classical BFT requires all honest nodes to agree on **exactly the same byte sequence**, whereas LLM Agent output is stochastic natural language and byte-level agreement is in principle impossible; so it lets each round produce one of three **typed results**—semantic_commit (signing a quantitative aggregated summary of the semantic core, fine-grained, high-information, higher risk), verdict_commit (signing only the pass/fail label, coarse-grained, low-information, safer), typed_abort (explicitly refusing to commit), all three sharing the same $\geq 2f+1$ distinct-signer certificate envelope. More importantly, H-CSC v2 does something rare in systems

papers: after re-checking with a stricter "steelman" baseline, it **actively withdrew** the v1 claim that "our filtered aggregation retains more honest content."

Based on this, this paper proposes a three-level typed finality design: rewrite the binary STOP/CONTINUE of P2 into `strong_stop` (semantic-core-level agreement, corresponding to `semantic_commit`), `weak_stop` (only verdict-label agreement, corresponding to `verdict_commit`), `abort` (not aligned, corresponding to `typed_abort` and triggering view change). This paper also turns the H-CSC v2 negative result into a red line: **do not use a learned encoder to replace rule checking**—in its adversarial experiments, the AUROC of the deterministic lexical consistency predicate (0.865–0.982) is significantly better than the learned semantic encoder (0.621–0.744). The three-level upgrade has not been implemented, and its safety/liveness properties, costs, and failure modes are all [RESULT NEEDED]; benefits can only be claimed after extending P2's existing property-test suite.

Keywords: typed finality; semantic BFT; stop decision; quorum; equivocation; steelman baseline; negative result; rule predicate

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** LLM committee output is stochastic natural language and cannot reach byte-level consensus; while the binary STOP/CONTINUE of v7.5.0 P2 compresses "how strong the semantic consensus is" into one switch, losing the safety information of the intermediate state "semantic core already aligned, only wording diverges."
 - **Core argument:** Stop decisions should be **graded by the degree of semantic agreement**, rather than binarized. The same $2f+1$ certificate envelope can bind different commit granularities: semantic-core agreement gives a strong stop, only-label agreement gives a weak stop, and non-agreement gives abort and turns to a view; grading lets the system gradient-match "wrong-stop risk" and "rework cost" rather than using one cut.
 - **Evidence threads:** (1) the typed commit object design of H-CSC v2 (paper-reported); (2) H-CSC v2 actively withdrawing "honest retention advantage" with a steelman baseline—the filtered aggregation is instead farther from the honest center ($5.32^\circ/5.48^\circ$ vs $4.27^\circ/4.32^\circ$), with honest retention 0.6475 vs 0.7325 against steelman M3, and the authors explicitly write "we withdraw it" (paper-reported); (3) deterministic lexical predicate AUROC 0.865–0.982 beats learned encoder 0.621–0.744 (paper-reported); (4) the 10 property tests and 24-order end-to-end of UDOS v7.5.0 P2 (cpu-proto / verified).
 - **Contributions:** (i) the formal design upgrading P2's binary stop decision into three-level typed finality; (ii) turning the H-CSC v2 self-withdrawn negative result into a UDOS design red line (rules as base, learning as supplement); (iii) a falsifiable property list for the three-level decision and pending [RESULT NEEDED] items; (iv) making the failure boundary of this proposal clear.
-

1 The Problem: What the Binary Stop Decision Loses

1.1 What P2 Got Right

The contribution of v7.5.0 P2 is to **reduce** the "safety/liveness" of classical BFT from natural-language proofs into a CI-able, regression-able property-test suite in the repo. It does not pursue generic state-

machine replication but serves one specific decision point—the acceptance committee voting on "whether a work order is done." Its skeleton is:

- Committee $V=\{v_1, \dots, v_n\}$, at most f members Byzantine, $n \geq 3f+1$;
- Each round members vote on a binary decision, quorum $q=2f+1$;
- Same-round equivocation (swing voting) members invalidate the whole round;
- More than f silence returns `NO_QUORUM` and triggers view change, rather than hanging.

At $n=7, f=2$ ($q=5$), 10 deterministic property tests (verified) cover illegal- n rejection, honest supermajority stop, Byzantine inability to force a stop, equivocation detection and whole-round invalidation, conflicting-quorum unreachability, silence triggering view change, and unknown-voter rejection; in the end-to-end integration (cpu-proto, single seed), the 7-person QA committee accepts all 24 work orders correctly under 2 Byzantine reverse-voting members. P2 also honestly states three boundaries: SHA-256 hash placeholder signatures are not cryptographically secure, network partitions and adaptive attackers are not modeled, and **semantic common-cause faults break through majority votes**.

1.2 The Link It Loses: The Decision Space Is Too Coarse

P2 limits the decision space to binary (STOP/CONTINUE), with the reason written clearly: the safety analysis of a binary quorum is clearest and fits the real terminal of "a work order is either accepted or reworked." This simplification is correct in the **byte-level** world—the replicated state machine of classical BFT votes on deterministic byte streams in the first place.

But it is not correct in the LLM world. Five conclusions a QA committee gives for the same work order may be:

1. **Semantic core aligned, wording diverges:** All five judge "pass," each writing their own reasons, with even the cited key numbers slightly differing;
2. **Labels aligned, reasons conflict:** All five check "pass," but three think "barely pass" and two think "fully pass," with acceptance confidence uneven;
3. **Completely not aligned:** Half pass and half fail.

A binary switch can only distinguish "whether someone gathered enough q STOP votes," and **cannot distinguish the risk levels of the three cases above**. Case 1 is actually already safe (semantic core aligned) yet shares the same "stop" with case 3; case 2 is the most dangerous—the label is captured by the majority while reasons split, and P2's property tests precisely point out this is the breeding ground for "semantic common-cause faults breaking through majority votes," but the binary decision gives it no separate outlet.

This is where H-CSC cuts in: **rather than voting on bytes, first decide "on what granularity of semantic product to vote."**

2 H-CSC's Solution: Classify First, Then Commit

2.1 Why Byte-Level Agreement Is Impossible

The premise of classical BFT is state-machine replication: all replicas execute **exactly the same** state transitions, so it suffices to vote on the same byte sequence. LLM Agents do not satisfy this premise—under the same prompt and temperature, two samplings produce two different natural-language

outputs. Requiring a committee to reach $2f+1$ agreement on "verbatim identical" text fails in principle (probability decays exponentially with length).

The core claim of H-CSC v2 (after retitling, Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides) is: **finality should be typed**—the protocol must first decide "what this round actually commits" and then sign. It gives three commit objects (paper-reported):

- `semantic_commit`: Signing a quantitative aggregated summary of the **semantic core**. Fine-grained, high-information, but to aggregate divergent natural language into a signable core, the aggregation process itself may introduce bias (higher risk);
- `verdict_commit`: Signing only the **label** (pass/fail). Coarse-grained, low-information, but the label is discrete and mechanically comparable, the safest;
- `typed_abort`: Explicitly **refusing to commit**, meaning this round produces no promise.

All three share the same $2f+1$ distinct-signer certificate envelope—that is, the safety skeleton of "how many endorse" is unchanged, and what changes is **what this certificate actually commits**. This design is very friendly to UDOS: it does not need to overturn P2's $n \geq 3f+1$, $2f+1$, and equivocation whole-round invalidation, but only needs to add, "after gathering the quorum," a step of "which level this quorum commits."

2.2 This Paper's Three-Level Mapping

Map the three H-CSC commit objects directly onto P2's stop decision (design proposal):

Three-level finality	Corresponding H-CSC object	Trigger condition (design)	Engineering meaning
strong_stop	<code>semantic_commit</code>	$q \geq 2f+1$ members agree on the semantic core (aggregated summary of acceptance points)	high-confidence completion, entering settlement/archiving
weak_stop	<code>verdict_commit</code>	$q \geq q$ members agree only on the pass label , but the semantic core does not reach agreement	low-confidence completion, leave a "barely pass" trace, sample-review next round
abort	<code>typed_abort</code>	Neither semantic core nor label reaches agreement, or a semantic common-cause fault sign is detected	turn to view change / add rounds, rather than hard-stopping or hard-reworking

Figure 1 Binary stop decision to three-level typed finality

Figure 1 The binary STOP/CONTINUE of v7.5.0 P2 compresses the semantic-consensus gradient into a switch; P16 upgrades it into `strong_stop` / `weak_stop` / `abort` three levels, with the same $2f+1$ certificate envelope binding different commit granularities. Conceptual illustration, not measured data.

The key benefit of this mapping is to give each of the three cases in Sec. 1.2 an outlet: case 1 lands in `strong_stop`, case 2 lands in `weak_stop` (and triggers next-round review rather than quietly high-confidence archiving), and case 3 lands in `abort` (turning to a view). **Binary P2 misreads case 2 as a safe "stop," while the three-level design makes it show up as "weak stop + review."**

3 H-CSC v2's Self-Withdrawal: This Paper's Most Important Negative Result

3.1 The Steelman Baseline and "we withdraw it"

Any design borrowing from H-CSC must first digest its v2 self-correction—this is exactly why this paper lists it as a dialogue object rather than a template.

H-CSC v1 once claimed: its "filtered aggregation" mechanism is better than a simple majority at retaining honest content. After re-checking with a **stricter steelman baseline** (i.e. giving the comparison side a stronger, fairer opponent rather than deliberately finding a weak baseline to set itself off), v2 found (paper-reported numbers, not independently rechecked):

- H3: The angle of the core aggregation from the honest center is **5.32° / 5.48°** for H-CSC, while the steelman baseline is **4.27° / 4.32°**—**H-CSC is instead farther from the honest center**;
- H4: Honest retention rate, under the v1 scope H-CSC 0.6475 vs weak baseline 0.4950 (looks dominant); but under the steelman M3 baseline it is **0.6475 vs 0.7325**—**H-CSC is instead lower**. The authors explicitly write "**we withdraw it**" (withdrawing the claim).

This withdrawal is extremely rare in systems papers, and its methodological value is greater than any of its positive conclusions: **a seemingly clever "learned filtered aggregation," under a fair baseline, not only brings no benefit but is worse**. It reminds UDOS: do not introduce a module because it "sounds smart"; every time a processing step is introduced, one must ask "who is its steelman opponent and did it really win." This is fully consistent with the restrained posture of P2 actively disclosing "the full-factorial contrast with single-point QA and naive majority votes has not been done."

3.2 Another Negative Result: The Learned Encoder Is Defeated by the Lexical Predicate

The second hard result of H-CSC v2 is likewise negative but with a clear direction (paper-reported, not independently rechecked): in adversarial scenarios, among the tools deciding "whether two natural-language outputs are semantically consistent," the **deterministic lexical consistency predicate** (mechanical comparison by an agreed set of keywords/phrases) has AUROC **0.865–0.982**, while the **learned CRSE semantic encoder** (using a neural network to encode text into a vector space and then compute similarity) has AUROC only **0.621–0.744**.

That is, on the task of "recognizing whether semantics are consistent," the more advanced-looking neural-network tool is significantly defeated by a simple rule predicate. This directly leads to P32 (rules over learning) and directly constrains this paper's design: **the step in three-level finality of "judging whether the semantic core is consistent" must first be based on a rule predicate, and the learned model can only be a supplement and must pass a steelman contrast before entering a safety-critical path**.

Figure 2 The two negative results of H-CSC v2

Figure 2 Left: in adversarial scenarios the deterministic lexical predicate AUROC (0.865–0.982) is significantly better than the learned CRSE encoder (0.621–0.744); right: under the steelman baseline the honest retention rate does not rise but falls, explicitly withdrawn in v2. Numbers are paper-reported, not independently rechecked.

4 Formal Design of Three-Level Typed Finality (Proposal)

4.1 Decision Function

Extended on P2's `decide(round_id)`. Suppose each round, besides voting STOP/CONTINUE, members also produce two signable products: L_i (verdict label, discrete) and C_i (comparable fingerprint of the semantic-core aggregated summary). After gathering $q=2f+1$ STOP votes, do one more grading judgment:

$$\text{level} = \begin{cases} \text{strong_stop}, & \text{if the } C_i \text{ of the } q \text{ votes are} \\ & \text{judged consistent by the rule predicate} \\ \text{weak_stop}, & \text{if the } L_i \text{ of the } q \text{ votes} \\ & \text{are consistent but } C_i \text{ are not} \\ \text{abort}, & \text{if } L_i \text{ do not reach agreement or a} \\ & \text{semantic common-cause sign is detected} \end{cases}$$

(This is design pseudocode; the concrete implementation of `cases` is rewritten with safety commands: the `strong_stop` / `weak_stop` / `abort` three branches are given by boolean combinations of rule predicates.) The safety skeleton is unchanged: equivocation invalidates the whole round, more than f silence returns `NO_QUORUM`, and abort triggers view change.

4.2 Relationship with P2's Three Properties

The three-level design **inherits** the three properties P2 has verified (verified):

- **Safety:** `strong_stop` and `abort` cannot be gathered at the same time, because their judgments of C_i are mutually exclusive;
- **Liveness:** `weak_stop` does not hang but gives "barely pass + next-round review"; `abort` turns to a view rather than waiting forever;
- **Illegal configuration rejection:** $n < 3f+1$ is still rejected at the constructor layer, independent of grading.

But it **adds** one property P2 does not have, which has not yet been verified (`[RESULT NEEDED]`):

- **H4-new (grading monotonicity):** If the degree of semantic-core agreement in a round is higher, the `strong_stop` share should not anomalously fall; under the steelman baseline, the "wrong-stop rate" of the three-level decision must be **no higher than** P2's binary decision. This comes directly from the H-CSC v2 lesson—any grading module must prove it is not a "worse filter" under a fair baseline.

4.3 Pending Implementation List

The following are all design proposals with no experimental support:

1. Add C_i semantic-core fingerprints and rule-predicate judgment in `udos7/topology/consensus.py`; `[RESULT NEEDED]`
2. Extend the 10 property tests into a regression suite covering three-level decisions; `[RESULT NEEDED]`
3. Design a steelman contrast (naive verdict majority vs three levels), measuring wrong-stop/rework rates; `[RESULT NEEDED]`
4. Verify that the "next-round review" of `weak_stop` does not introduce unbounded loops (liveness); `[RESULT NEEDED]`

5 Dross / Failure Boundaries / Falsifiable Conditions

Dross (what not to do):

1. **Do not copy H-CSC's learned aggregation as safety-critical.** Its steelman contrast proves it is worse (5.32° vs 4.27°; honest retention 0.6475 vs 0.7325), and this paper only takes its "typed commit" idea, not its learned-aggregation implementation.
2. **Do not treat strong_stop as "absolutely safe."** The $\$2f+1\$$ certificate only guarantees "no $\$>f\$$ Byzantine," not that semantic common-cause faults do not break through—P2 has stated this boundary, and the three-level design does not remove it.
3. **Do not misread "three levels" as "all three levels can settle."** weak_stop is "leave a trace + review," not high-confidence archiving; if engineering also settles weak_stop directly, it degenerates into a binary decision with labels.

Failure boundaries:

- When the rule predicate for $\$C_i\$$ is designed too loosely (misjudging wording divergence as semantic agreement), strong_stop will be inflated and the wrong-stop rate rises;
- When a semantic common-cause fault occurs (all QA misled by the same point), the "semantic-core agreement" of strong_stop may itself be **collusive agreement**, and the certificate cannot distinguish "honest agreement" from "induced agreement";
- Three-level judgment introduces an extra round of fingerprint aggregation, costing more than binary decisions, and at very large scale fan-in must be evaluated (coupled with P1 topology complexity).

Falsifiable conditions:

- If under the steelman baseline the wrong-stop rate of the three-level decision is **not lower than** P2's binary decision, H4-new is overturned and this proposal should be rolled back;
- If the rule predicate's AUROC on a new attack family falls below the learned encoder (the current 0.865–0.982 vs 0.621–0.744 advantage reverses), the "rules as base" red line must be re-evaluated;
- If the review of weak_stop in actual measurement introduces significant unbounded loops, the liveness hypothesis of the three-level design fails.

6 Conclusion

v7.5.0 P2 makes the stop decision into a falsifiable, CI-able binary committee; H-CSC v2 reveals the semantic-consensus gradient lost by binarization and, with a rare self-withdrawal, warns that "a learned clever module may be worse under a fair baseline." This paper's stance is restrained: **only take the typed-commit idea, upgrading P2's STOP/CONTINUE into strong_stop / weak_stop / abort three levels; the step of judging semantic consistency is based on a rule predicate with the learned model filling in, and a steelman contrast is mandatory.** Three-level finality is a design proposal whose benefits and failure modes are all [RESULT NEEDED]—before P2's property-test suite is extended and the steelman contrast runs through, this paper claims no actual improvement.

Replacing "voting on bytes" with "first deciding on what granularity of semantic promise to vote" is a step BFT in the LLM era must cross; but when crossing this step, use the honesty with which H-CSC v2 itself withdrew its conclusion to examine every newly introduced step.

References

Classical literature

1. Castro, M., & Liskov, B. (1999). Practical Byzantine Fault Tolerance. OSDI 1999. (the classical skeleton of $3f+1$, $2f+1$ quorum, view change)
2. Yin, M., Malkhi, D., Reiter, M. K., et al. (2019). HotStuff: BFT Consensus in the Lens of Blockchain. PODC 2019. arXiv:1803.05069. (linear-communication-complexity BFT, chained view change)
3. Wu, Q., Bansal, G., Zhang, Y., et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155. (conversable multi-agent framework, whose "conversations without termination conditions easily dead-loop" is one motivation of this paper)

2026 frontier materials (existence verified; numbers paper-reported, not independently rechecked)

1. H-CSC author team (2026). Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides (v2; v1 original title Hierarchical Certified Semantic Commitment...). arXiv:2606.07316v2. (typed commit objects semantic_commit / verdict_commit / typed_abort, $2f+1$ certificate envelope; actively withdrawing "honest retention advantage" under the steelman baseline—aggregation distance $5.32^\circ/5.48^\circ$ vs $4.27^\circ/4.32^\circ$, honest retention 0.6475 vs 0.7325; lexical predicate AUROC 0.865–0.982 beats learned encoder 0.621–0.744)

UDOS paper volume (evidence grades marked in text)

1. UDOS v7.5.0 P2 "BFT-lite for Multi-Agent Stop Decisions: Safety Verification of the $n \geq 3f+1$ Committee, Equivocation Whole-Round Invalidation, and View Change" (`udos7/topology/consensus.py`, 10 property tests verified; $n=7, f=2$, 24-order end-to-end cpu-proto).
2. UDOS v7.6.0 P26 "Weighted BFT and Reputation: The Weight Direction of Equal-Weight $2f+1$ ", P32 "Rules over Learning: Predicate Robustness Boundaries in Adversarial Scenarios" (sister papers).

Evidence Discipline and Reproduction Notes

- This paper is a design-proposal paper with no new experiment; all H-CSC numbers (aggregation distance, honest retention rate, AUROC intervals) are **paper-reported, not independently rechecked**, and its v2 has actively withdrawn v1's "honest retention advantage" claim. This paper strictly adopts the v2 scope and does not use v1 positive conclusions.
- All UDOS-side numbers go back to v7.5.0 P2: the 10 property tests are verified, and the 24-order end-to-end at $n=7, f=2$ is a cpu-proto single-seed pilot point estimate; the three boundaries of P2—SHA-256 hash placeholder signatures are not cryptographically secure, network partitions and adaptive attackers are not modeled, and semantic common-cause faults break through majority votes—**still hold** for this paper's three-level design.
- All benefits of three-level typed finality, H4-new monotonicity, and the `weak_stop` unbounded-loop risk are marked `[RESULT NEEDED]`, claimable only after extending the existing `consensus.py` and running through the steelman contrast; ≥ 30 -seed reproduction is needed before submission.